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Quantum Computing

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9 november 2026

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1

Complex linear algebra

1.1 Complex numbers

Complex number

A **Complex number** z is defined as:

$$z = a + ib$$

where $b \in \mathbb{R}$.

From a complex number we can define:

- Coniugate: $\overline{z} = \overline{a + ib} = a - ib$
- Modulus: $|z| = \sqrt{a^2 + b^2}$

Given a number z and its modulus $|z|$, we can find an angle $\theta \in [-\pi, \pi]$ such that:

$$\cos\theta = \frac{a}{\sqrt{a^2 + b^2}} = \frac{a}{|z|}$$
$$\sin\theta = \frac{b}{\sqrt{a^2 + b^2}} = \frac{b}{|z|}$$

So, we can write:

$$z = |z|(\cos\theta + i\sin\theta)$$

We also define the exponential of an imaginary number:

$$e^{i\theta} = \cos\theta + i\sin\theta \implies z = |z|e^{i\theta}$$

1.2 Vectors and Matrices

A vector n is list of complex numbers, for which we can define a norm:

$$||v|| = \sqrt{|v_1|^2 + \cdots + |v_n|^2} = \sqrt{\sum_{i=1}^{n} |v_i|^2}$$

Example:
Given the vector:

$$v = [4, i, 2 + 3i]$$

Its norm is:

$$||v|| = \sqrt{16 + 1 + 13} = \sqrt{30}$$

A matrix is a table of complex number, for which the multiplication is not commutative. The multiplication between a matrix A with size $m \times n$ and a matrix B with size $n \times q$ has size $m \times q$. The result matrix C has in every cell:

$$C_{i,j} = \sum_{k=0}^{n} A_{i,k} \cdot B_{k,j}$$

Example:
Given the matrices:

$$A = \begin{bmatrix} 3 & i \\ 1+i & 0 \\ -4 & 2+3i \end{bmatrix} \quad B = \begin{bmatrix} 3 & 1 \\ 0 & 0 \end{bmatrix}$$

The multiplication $A \cdot B$ is:

$$C = \begin{bmatrix} 9 & 3 \\ 3+3i & 1+i \\ -12 & -4 \end{bmatrix}$$

1.2.1 Eigenvectors and Eigenvalues

A vector v is an **eigenvector** with **eigenvalue** λ for a matrix A if:

$$A \cdot v = \lambda \cdot v$$

We can also write:

$$A \cdot v - \lambda \cdot v = 0 \iff (A - \lambda I) \cdot v = 0$$

This equation has non-zero solutions only if:

$$\det(A - \lambda I) = 0$$

In general v and λ could be complex, but quantum mechanics uses matrices that always have eigenvalues $\lambda \in \mathbb{R}$.

2

Introduction

2.1 Probabilistic algorithms

In normal programming, an algorithms produce a fixed output for a given input. With probabilistic algorithms there is a finite set of possible outputs $\{o_1, \ldots, o_N\}$ for a given input. Each outputs o_i has a probability of p_i to be produced, where:

$$\sum_{k=1}^{N} p_k = 1$$

Classical algorithms can have probabilistic outputs based on pseudo-random number generator (PRNG), that produce a sequance of numbers that appear random but after many elements the sequence repeats. In quantum algorithms there is no PRNG, but real randomicity.

2.2 Qubit

Qubit

A **qubit** is the quantum equivalent of a bit, where the basis states are:

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

The true state of a qubit ψ is a **superposition** defined as:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

where $\alpha, \beta \in \mathbb{C}$ are called **probability amplitudes** and $|\alpha|^2 + |\beta|^2 = 1$.

The notation $|\psi\rangle$ is called "ket".

Example:

Given the vector:

$$v = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}$$

The corresponding qubit is:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} \left(|0\rangle + |1\rangle \right)$$

The true state of a qubit cannot be observed, but only measured. Measuring a qubit makes it collapse into $|0\rangle$ or $ket1$ with the probabilities $|\alpha|^2$ and $|\beta|^2$. We write this property as:

$$\alpha\,|0\rangle + \beta\,|1\rangle \begin{cases} |0\rangle & @\ |\alpha|^2 \\ |1\rangle & @\ |\beta|^2 \end{cases}$$

where @ is called "at".
For any complex number z such that $|z| = 1$, we have that $|z\alpha| = |\alpha|$, so when measuring we have that:

$$z\alpha\,|0\rangle + z\beta\,|1\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

because the probability is based on the amplitudes squared.

2.2.1 Bloch sphere

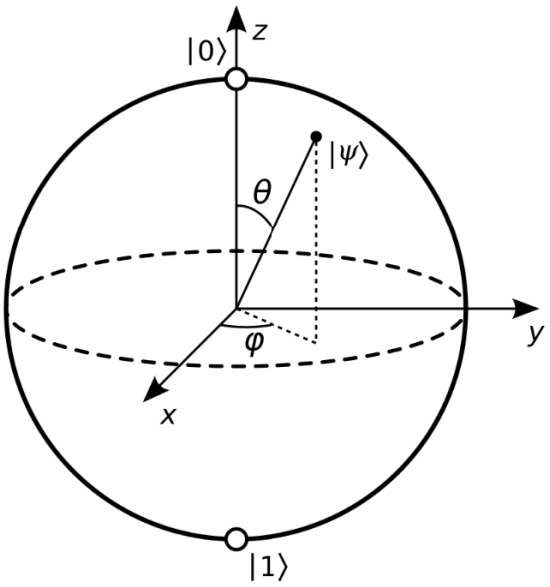
When can picture a qubit using the **Bloch sphere**. Considering a qubit:

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle$$

Rewriting it using polar coordinates:

$$|\psi\rangle = e^{i\gamma}\left(\cos\frac{\theta}{2}\,|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}\,|1\rangle\right)$$

Since $|e^{i\gamma}| = 1$ we can remove it, representing the qubit based on two angles $\theta \in [0, \pi]$ and $\varphi \in [0, 2\pi)$.



There is no Bloch sphere equivalent for systems composed by two or more qubits.

2.3 Qubit operations

Qubits transformations are linear, so they can be represented as a matrix.

Pauli matrices

The **Pauli matrices** are 2×2 matrices that represent operations:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

These are called the X, Y and Z gate respectively.

The result of applying this gates on the two basis states $|0\rangle$ and $|1\rangle$ are:

- X gate (also called NOT):

$$\sigma_x |0\rangle = |1\rangle \qquad \sigma_x |1\rangle = |0\rangle$$

- Y gate:

$$\sigma_y |0\rangle = i |1\rangle \qquad \sigma_y |1\rangle = -i |0\rangle$$

- Z gate:

$$\sigma_z |0\rangle = |0\rangle \qquad \sigma_z |1\rangle = - |1\rangle$$

Another important transformation is the **Hadamard gate**, represented by the following matrix:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

This gate can map a classical state into a superposition:

$$H |0\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \left[\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right] = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

If we measure it we will have a true random bits generator:

$$\frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) \begin{cases} |0\rangle & @ \ \frac{1}{2} \\ |1\rangle & @ \ \frac{1}{2} \end{cases}$$

We define the result of applying the Hadamard gate on the two classical state with special names:

$$H |0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle) = |+\rangle \qquad H |1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle) = |-\rangle$$

To be an allowed quantum transformation a matrix must preserve the norm of the input vector, so it must be **unitary**. Since $|\alpha|^2 + |\beta|^2 = 1$, the norm of every vector v must be $||v|| = 1$ for any qubit state. An unitary matrix preserve the probabilities and is reversible, so from the output we can recover the input.

2.4 Multiple qubits

In classical computers we can describe n bits using an vector of elements and all the possible combination are obtained by performing the cartesian product:

$$\mathcal{B} = \{0, 1\} \implies \mathcal{B} \times \mathcal{B} = \{00, 01, 10, 11\}$$

The lenght of the vector grows linear in respect to n and each element is independent from the others.
To represent all the state of two qubits we need to use the **tensor product**. Given two qubits, the tensor product of their state vectors:

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix}\alpha_0\\ \alpha_1\end{pmatrix} \otimes \begin{pmatrix}\beta_0\\ \beta_1\end{pmatrix} = \begin{pmatrix}\alpha_0\beta_0\\ \alpha_0\beta_1\\ \alpha_1\beta_0\\ \alpha_1\beta_1\end{pmatrix} = \begin{pmatrix}\alpha_0 |\phi\rangle\\ \alpha_1 |\phi\rangle\end{pmatrix} = |\psi\phi\rangle$$

This product can also be written by using the **diagonal (or computational) basis vectors**, that are vectors with all 0 and a 1 in a single position i, called e_i.

$$\alpha_0\beta_0 \underbrace{\begin{pmatrix}1\\0\\0\\0\end{pmatrix}}_{e_0} + \alpha_0\beta_1 \underbrace{\begin{pmatrix}0\\1\\0\\0\end{pmatrix}}_{e_1} + \alpha_1\beta_0 \underbrace{\begin{pmatrix}0\\0\\1\\0\end{pmatrix}}_{e_2} + \alpha_1\beta_1 \underbrace{\begin{pmatrix}0\\0\\0\\1\end{pmatrix}}_{e_3} = \alpha_0\beta_0 |00\rangle + \alpha_0\beta_1 |01\rangle + \alpha_1\beta_0 |10\rangle + \alpha_1\beta_1 |11\rangle$$

We can see that each vector e_i is equal to the vector state of $|i\rangle$ in which we consider the binary representation of i. So, the previous result can also be written as a sum:

$$\sum_{i,j\in\{0,1\}} \alpha_i\beta_j |ij\rangle$$

Example:
Given two qubits:

$$|\psi\rangle = |\phi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

Their tensor product is:

$$|\psi\rangle \otimes |\phi\rangle = \begin{pmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{pmatrix} \otimes \begin{pmatrix}\frac{1}{\sqrt{2}}\\ \frac{1}{\sqrt{2}}\end{pmatrix} = \begin{pmatrix}\frac{1}{2}\\ \frac{1}{2}\\ \frac{1}{2}\\ \frac{1}{2}\end{pmatrix} = \frac{1}{2}(|00\rangle + |01\rangle + |10\rangle + |11\rangle)$$

2.4.1 Entanglement

In the example above the two qubits can be separated and decomposed into two valid qubit states.
Since any qubit written:

$$\alpha_0 |00\rangle + \alpha_1 |01\rangle + \alpha_2 |10\rangle + \alpha_3 |11\rangle \quad \text{with } \sum_i |\alpha_i|^2 = 2$$

is a valid quantum state for a 2-qubit system we can have a valid qubit such as:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

It's not possible to separate this state in two different quantum states for which their tensor product is $|\psi\rangle$. This would require:

$$|a\rangle \otimes |b\rangle = \alpha_0\beta_0 |00\rangle + \alpha_0 v\beta_1 |01\rangle + \alpha_1\beta_0 |10\rangle + \alpha_1\beta_1 |11\rangle = \begin{cases} \alpha_0\beta_0 = \alpha_1\beta_1 = 0 \\ \alpha_0\beta_1 = \alpha_1\beta_0 = \frac{1}{\sqrt{2}} \end{cases}$$

The product $\alpha_0\beta_0 = 0$ implies that at least one of the two is 0, but this means also that at least one of $\alpha_0\beta_1$ or $\alpha_1\beta_0$ is 0.
This state is **entangled**, it means that it doesn't describe two qubit states but the entire 2-qubit system as a whole. This entangled states are called the **EPR** states and when measuring them we see that:

$$|\psi\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) \begin{cases} |01\rangle & @ \frac{1}{2} \\ |10\rangle & @ \frac{1}{2} \end{cases}$$

This pairs also have a strange behaviour when physically separated. Since we cannot measure $|00\rangle$ or $|11\rangle$, measuring one of the two qubit is enough to know the entire system, also when the two qubits are kilometers apart. In fact is proved that if we take the two qubits apart and measure one of the two, the other will also collapse its state.

2.4.2 Mutiple qubits gates

The basic gate for two qubits its the CNOT gate, where the first bit is the **control** and the second one the **target**. The target bit is flipped when the control is 1, so the truth table is:

a	b	CNOT(a,b)
0	0	0
0	1	1
1	0	1
1	1	0

Logically we have that:

$$\text{CNOT}(a, b) = a \oplus b$$

Since we want the computation to be invertible we need the value of a to recover the value of b, so the quantum CNOT behaves as follows:

$$\text{CNOT}(|a\rangle \otimes |b\rangle) = |a\rangle \otimes |a \oplus b\rangle = |a, a \oplus b\rangle$$

The matrix that represent this gate is:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

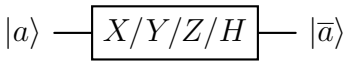
2.5 Quantum circuit

In quantum computing we program using gates and circuits. Each circuit U must be unitary, that means:

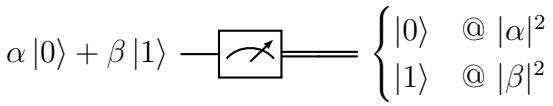
$$UU^\dagger = U^\dagger U = I$$

Where $U^\dagger$ is the **adjoint** of U, given by $U^\dagger_{i,j} = \overline{U_{j,i}}$.
The notation used for the gates seen before is:

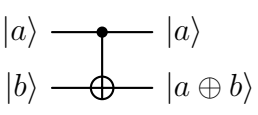
- X, Y, Z and H gates: single bit with the correspondig letter

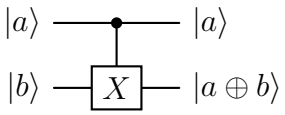


- Measurment gate: the double wire after means that is passing classical bits, not quantun ones



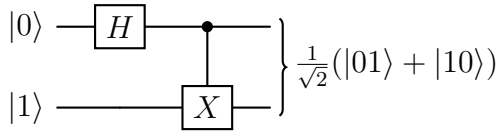
- CNOT gate: the wire with the dot is the control and the one with the gato or xor is the target





2.5.1 Creating EPR pairs

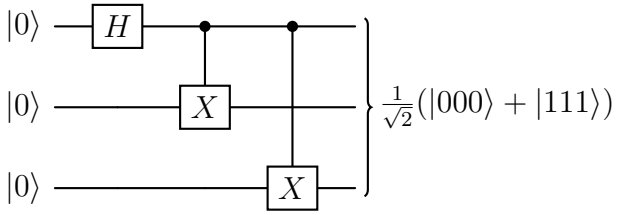
There are different circuits capable of creating EPR pairs, one of the common is composed by H and CNOT gates.



We can evaluate the circuit by sequentially crossing the gates from left to right:

$$|0\rangle \otimes |1\rangle \xrightarrow{H} \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \otimes |1\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) \xrightarrow{\text{CNOT}} \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

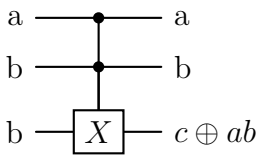
Another gate that creates EPR pairs, but with 3 qubits is the GHZ (Greenberger, Horne, Zeilinger) gate:



Tecnically all quantum circuits starts with all $|0\rangle$ as inputs, but we can simply add a NOT gate to change it to $|1\rangle$.

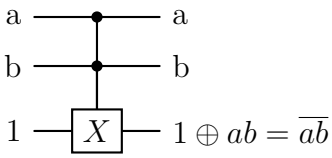
2.5.2 Reversible classical computing

Using a specific quantum gate, called the **Toffoli** gate (also called CCNOT), we make classical computing reversible.

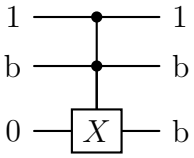


With this gate we can create the NAND and FANOUT gate, that means that any boolean circuit can be implemented only with Toffoli gate, making it reversible.

NAND gate



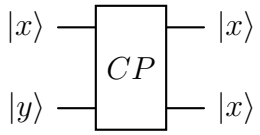
FANOUT gate



2.6 Cloning and Teleportation

2.6.1 No cloning theorem

A copy operation takes a source variable to be copied and a destination variable where the copy is applied. As a circuit we could write:



We need a variable $|y\rangle$ that is a "white paper" where the content of $|x\rangle$ can be copied, but this cannot exists.

No cloning theorem

There is no operation CP, such that:

$$\exists\, |y\rangle\, |\forall\, |x\rangle \;\; \mathrm{CP}(|x\rangle \otimes |y\rangle) = |x\rangle \otimes |x\rangle$$

Proof by contradiction:

For the proof the $|\rangle$ notation will be avoided.

CP must work on superposed states, so:

$$\exists y | \forall x, a \;\; \mathrm{CP}(\underbrace{(x+a) \otimes y}_{(x \otimes y)+(a \otimes y)}) = (x+a) \otimes (x+a)$$

CP must be linear:

$$\exists y | \forall x, a \ \mathrm{CP}(x \otimes y) + \mathrm{CP}(a \otimes y) = (x + a) \otimes (x + a)$$

Since CP copies its input:

$$\forall x, a \ (x \otimes x) + (a \otimes a) = (x + a) \otimes (x + a) = (x \otimes x) + (x \otimes a) + (a \otimes x) + (a \otimes a)$$

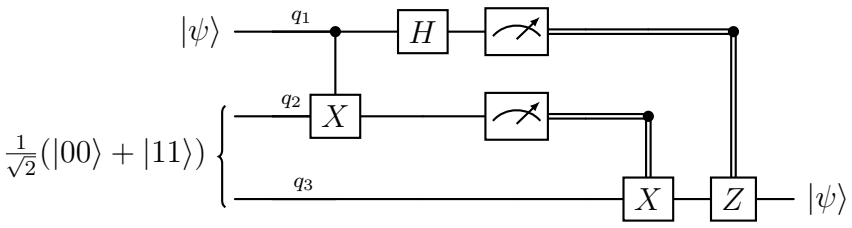
But we can prove with a simple example like $x = |0\rangle$ and $a = |1\rangle$ that this equation is not true.

$$\left.\begin{array}{l} x = |0\rangle \\ a = |1\rangle \end{array}\right\} \implies (x \otimes x) + (a \otimes a) = |00\rangle + |11\rangle \neq (x \otimes x) + (x \otimes a) + (a \otimes x) + (a \otimes a) = |00\rangle + |01\rangle + |10\rangle + |11\rangle$$

This means that we cannot clone perfectly, but **imperfect cloning** with high probability is possible.

2.6.2 Quantum teleportation

Even if we cannot perfectly clone a qubit, we can teleport it perfectly fine.
Suppose Alice has a qubit $|\psi\rangle$ and she wants to send it to Bob, this can be achieved by using 2 qubits and 2 classical bits. The Quantum teleportation circuit is the following:



We can give a pseudocode for the algorithm:

Algorithm: Quantum Teleportation

```
def Teleport(q1, q2, q3):
    q1, q2, q3=|ψ⟩ 1/√2(|00⟩ + |11⟩)
    q1, q2=CNOT(q1, q2)
    q1=H(q1)
    b1, b2=Measure(q1, q2)
    b2, q3=CNOT(b2, q3)
    b1, q3=CZ(b1, q3)
    return q3
```

We can prove that the algorithm works by executing the operations done by the code step by step, starting from a random qubit ψ.

$$|\psi\rangle \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (\alpha |0\rangle + \beta |1\rangle)\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = \frac{1}{\sqrt{2}} [\alpha |0\rangle (|00\rangle + |11\rangle) + \beta |1\rangle (|00\rangle + |11\rangle)]$$

Applying the CNOT on q_1 and q_2 we obtain:

$$\frac{1}{\sqrt{2}} [\alpha |0\rangle (|00\rangle + |11\rangle) + \beta |1\rangle (|10\rangle + |01\rangle)]$$

Next, we apply the Hadamard gate on q_1:

$$\frac{1}{\sqrt{2}}\left[\alpha\frac{1}{\sqrt{2}}(|0\rangle+|1\rangle)(|00\rangle+|11\rangle)+\beta\frac{1}{\sqrt{2}}(|0\rangle-|1\rangle)(|10\rangle+|01\rangle)\right]=$$
$$\frac{1}{2}\left[\alpha(|000\rangle+|011\rangle+|100\rangle+|111\rangle)+\beta(|010\rangle+|001\rangle-|110\rangle-|101\rangle)\right]=$$
$$\frac{1}{2}\left[|00\rangle\left(\alpha|0\rangle+\beta|1\rangle\right)+|01\rangle\left(\alpha|1\rangle+\beta|0\rangle\right)+|10\rangle\left(\alpha|0\rangle-\beta|1\rangle\right)+|11\rangle\left(\alpha|1\rangle-\beta|0\rangle\right)\right]$$

Now we have four possible outcomes based on the measurment done by Alice. Given the measured bits Bob need to apply different operation on its remaining qubit, to obtain the initial qubit $|\psi\rangle\ (\alpha|0\rangle+\beta|1\rangle)$.

Alice bits	Bob qubit	Operations
00	$\|\phi\rangle=\alpha\|0\rangle+\beta\|1\rangle$	$I\|\phi\rangle=\|\psi\rangle$
01	$\|\phi\rangle=\alpha\|1\rangle+\beta\|0\rangle$	$X\|\phi\rangle=\|\psi\rangle$
10	$\|\phi\rangle=\alpha\|0\rangle-\beta\|1\rangle$	$Z\|\phi\rangle=\|\psi\rangle$
11	$\|\phi\rangle=\alpha\|1\rangle-\beta\|0\rangle$	$XZ\|\phi\rangle=\|\psi\rangle$

Since Alice has measured her qubit, the qubit collapsed and she lost it forever, so now only Bob has that qubit.

3

Quantum parallelism

Given any boolean function $f : \mathcal{B}^n \to \mathcal{B}$, we can show that the function:

$$U_f = (x, y) \in (\mathcal{B}^n, \mathcal{B}) \to (x, y \oplus f(x))$$

Is reversible and unitary, so its a valid quantum gate.

Example:

For a 1-bit function we have the truth table:

		$y \oplus f(x)$		
x	y	x	$f(x) = 0$	$f(x) = 1$
0	0	0	0	1
0	1	0	1	0
1	0	1	0	1
1	1	1	1	0

So, we have that:

$$\begin{cases} U_f(x, 0) = (x, f(x)) \\ U_f(x, 1) = (x, \neg f(x)) \end{cases}$$

Since the function U_f can be implemented unitarly, we can apply it to qubits, obtaining:

$$\begin{cases} U_f \left| x \otimes 0 \right\rangle = \left| x \otimes f(x) \right\rangle \\ U_f \left| x \otimes 1 \right\rangle = \left| x \otimes \neg f(x) \right\rangle \end{cases}$$

Example:

Given $\left| x \right\rangle, \left| y \right\rangle$ as:

$$\left| x \right\rangle = \frac{1}{\sqrt{2}}(\left| 0 \right\rangle + \left| 1 \right\rangle) \qquad \left| y \right\rangle = \left| 0 \right\rangle$$

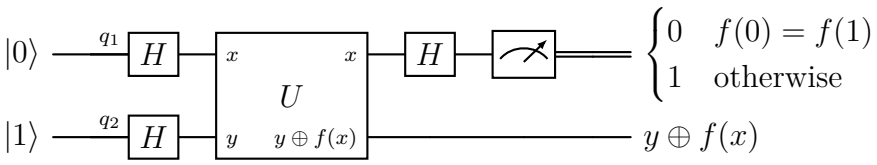
We can apply U_f to obtain:

$$U_f \left(\frac{1}{\sqrt{2}}(\left| 0 \right\rangle + \left| 1 \right\rangle) \otimes \left| 0 \right\rangle \right) = \frac{1}{\sqrt{2}}(U_f \left| 00 \right\rangle + U_f \left| 10 \right\rangle) = \frac{1}{\sqrt{2}}(\left| 0 \otimes f(0) \right\rangle + \left| 1 \right\rangle \otimes f(1))$$

So $f(0)$ and $f(1)$ are computed in **parallel**.

3.1 Deutsch algorithm

Given a bolean function $f : \mathcal{B} \to \mathcal{B}$ we want to compute the value of $f(0) \oplus f(1)$, to know if $f(0)$ is equal to $f(1)$. In classical computing we need 2 evaluation of the function f, but in quantum we can use the **Deutsch algorithm** that needs only one evaluation.



The code for the algorithm is the following:

Algorithm: Deutsch algorithm

```
def Deutsch(f, q1, q2):
    q1, q2=|01⟩
    q1, q2=H⊕H(q1, q2)
    q1, q2=Uf(q1, q2)
    q1=H(q1)
    b=Measure(q1)
    return b
```

Before proving the algorithm we need to know how U_f works on a random state a.

Application of U_f

Given a boolean function f, for any basis state $a \in \mathcal{B}^n$, we have that:

$$U_f\left(|a\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right) = (-1)^{f(a)}|a\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = (-1)^{f(a)}|a-\rangle$$

Proof:

$$U_f\left(|a\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right) = \frac{1}{\sqrt{2}}U_f(|a0\rangle - |a1\rangle) = \frac{1}{\sqrt{2}}(|af(a)\rangle - |a\neg f(a)\rangle)$$

We need to divide the result based on the value of $f(a)$:

$$\frac{1}{\sqrt{2}}(|af(a)\rangle - |a\neg f(a)\rangle) = \begin{cases} \frac{1}{\sqrt{2}}(|a0\rangle - |a1\rangle) & f(a) = 0 \\ \frac{1}{\sqrt{2}}(|a1\rangle - |a0\rangle) & f(a) = 1 \end{cases} = \frac{1}{\sqrt{2}}(-1)^{f(a)}|a\rangle\,(|0\rangle - |1\rangle) = (-1)^{f(a)}|a-\rangle$$

Now, from the starting state $|01\rangle$ we can execute the operations of the algorithm in order.

$$|01\rangle \xrightarrow{H\otimes H(q_1,q_2)} |+-\rangle = \frac{1}{\sqrt{2}}|0-\rangle + \frac{1}{\sqrt{2}}|1-\rangle \xrightarrow{U_f(q_1,q_2)} \frac{1}{\sqrt{2}}\left((-1)^{f(0)}|0-\rangle + (-1)^{f(1)}|1-\rangle\right)$$

Based on the value of $f(0)$ and $f(1)$ we can separate the result:

$$\begin{cases} \frac{1}{\sqrt{2}}(|0-\rangle + |1-\rangle) & f(0)=0, f(1)=0 \\ \frac{1}{\sqrt{2}}(-|0-\rangle - |1-\rangle) & f(0)=1, f(1)=1 \\ \frac{1}{\sqrt{2}}(|0-\rangle - |1-\rangle) & f(0)=0, f(1)=1 \\ \frac{1}{\sqrt{2}}(-|0-\rangle + |1-\rangle) & f(0)=1, f(1)=0 \end{cases} \implies \begin{cases} \pm\frac{1}{\sqrt{2}}(|0-\rangle + |1-\rangle) & f(0)=f(1) \\ \pm\frac{1}{\sqrt{2}}(|0-\rangle - |1-\rangle) & f(0)\neq f(1) \end{cases}$$

$$\implies \begin{cases} \pm\frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle) + \frac{1}{\sqrt{2}}(|10\rangle - |11\rangle)\right) = \pm|+-\rangle \\ \pm\frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle) + \frac{1}{\sqrt{2}}(|10\rangle + |11\rangle)\right) = \pm|--\rangle \end{cases}$$

Now we apply the Hadamard gate only on the first qubit:

$$\begin{cases} \pm\frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle) + \frac{1}{\sqrt{2}}(|10\rangle - |11\rangle)\right) = \pm|+-\rangle \\ \pm\frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|00\rangle - |01\rangle) + \frac{1}{\sqrt{2}}(|10\rangle + |11\rangle)\right) = \pm|--\rangle \end{cases} \implies \begin{cases} |0-\rangle & f(0)=f(1) \\ |1-\rangle & f(0)\neq f(1) \end{cases}$$

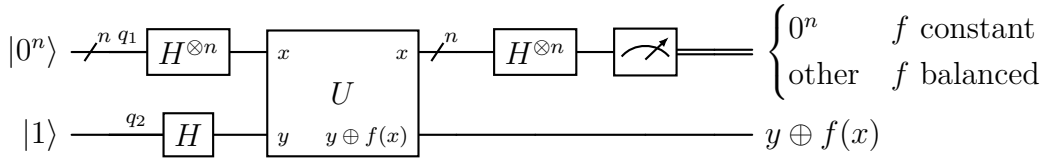
Measuring the first qubit, we will get 0 if $f(0)=f(1)$ and 1 if $f(0)\neq f(1)$, knowing also the result of $f(0)\oplus f(1)$.

3.2 Deutsch-Jozsa algorithm

Any boolean function $f:\mathcal{B}^n\to\mathcal{B}$ can be:

- Constant:
$$\exists b\in\mathcal{B}|\forall x\ f(x)=b$$
- Balanced:
$$\sum_{x\in\mathcal{B}^n} f(x) = \frac{2^n}{2} = 2^{n-1}$$

Given a function f that its either constant or balanced, we want to know wheter f is constant or balanced. The problem in classical computing needs at most $2^{n-1}+1$ evaluation of the function f, but in quantum the computation can be done with only one evaluation, having an exponential speedup.



The code for the algorithm is the following:

Algorithm: Deutsch-Jozsa algorithm

```
def Deutsch_Jozsa(f, q1, q2):
    q1, q2=|0^n 1⟩
    q1, q2=H^⊗n ⊕ H(q1, q2)
    q1, q2=U_f(q1, q2)
    q1=H^⊗n(q1)
    b=Measure(q1)
    return b
```

Also in this case before proving the algorithm we need to know how a gate works, specifically how $H^{\otimes n}$ works.

Application of $H^{\otimes n}$

For any $x \in \mathcal{B}^n$, we have that:

$$H^{\otimes n} |x\rangle = \frac{1}{\sqrt{2^n}} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a} |a\rangle$$

Where $x \cdot a$ is defined as:

$$x \cdot a = \left(\sum_{i=1}^{n} x_i a_i \right) \mod 2$$

Proof:

The result of a single Hadamard gate can be written as:

$$\left.\begin{aligned} H|0\rangle = |+\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = \tfrac{1}{\sqrt{2}} \sum_{b \in \mathcal{B}} \overbrace{(-1)^{0 \cdot b}}^{1} |b\rangle \\ H|1\rangle = |-\rangle = \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = \tfrac{1}{\sqrt{2}} \sum_{b \in \mathcal{B}} (-1)^{1 \cdot b} |b\rangle \end{aligned}\right\} \implies \forall a \in \mathcal{B}\ H|a\rangle = \frac{1}{\sqrt{2}} \sum_{b \in \mathcal{B}} (-1)^{a \cdot b} |b\rangle$$

So, the tensor product of n Hadamard gate for $x \in \mathcal{B}^n$ will be:

$$\begin{aligned} H^{\otimes n} |x\rangle = \bigotimes_{i=1}^{n} H |x_i\rangle = \bigotimes_{i=1}^{n} \left(\frac{1}{\sqrt{2}} \sum_{b \in \mathcal{B}} (-1)^{x_i \cdot b} |b\rangle \right) = \frac{1}{\sqrt{2^n}} \bigotimes_{i=1}^{n} (|0\rangle + (-1)^{x_i} |1\rangle) \\ = \frac{1}{\sqrt{2^n}} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a} |a\rangle \end{aligned}$$

Now we can follow the instruction of the algorithm starting with the state $|0^n 1\rangle$.

$$|0^n 1\rangle \xrightarrow{H^{\otimes n} H(q_1, q_2)} \left(\frac{1}{\sqrt{2^n}} \sum_{x \in \mathcal{B}^n} \overbrace{(-1)^{0^n \cdot x}}^{1} |x\rangle \right) \otimes |-\rangle \xrightarrow{U_f(q_1, q_2)} \left(\frac{1}{\sqrt{2^n}} \sum_{x \in \mathcal{B}^n} (-1)^{f(x)} |x\rangle \right) \otimes |-\rangle$$

After applying U_f the second qubit is never used anymore, so we can erase it and continue the computation with only q_1.

$$\begin{aligned} \frac{1}{\sqrt{2^n}} \sum_{x \in \mathcal{B}^n} (-1)^{f(x)} |x\rangle \xrightarrow{H^{\otimes n}(q_1)} \frac{1}{\sqrt{2^n}} \sum_{x \in \mathcal{B}^n} (-1)^{f(x)} \left(\frac{1}{\sqrt{2^n}} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a} |a\rangle \right) \\ = \frac{1}{2^n} \sum_{x \in \mathcal{B}^n} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a + f(x)} |a\rangle \end{aligned}$$

The two sum are not dependant one to the other so we can switch the order:

$$\frac{1}{2^n} \sum_{x \in \mathcal{B}^n} \sum_{a \in \mathcal{B}^n} (-1)^{x \cdot a + f(x)} |a\rangle = \frac{1}{2^n} \sum_{a \in \mathcal{B}^n} |a\rangle \sum_{x \in \mathcal{B}^n} (-1)^{x \cdot a + f(x)}$$

When measuring q_1 the probability of getting 0^n is computed by setting $a = 0^n$ in the formula above.

$$\frac{1}{2^n} \sum_{x \in \mathcal{B}^n} (-1)^{\overbrace{x \cdot a}^{0} + f(x)} = \frac{1}{2^n} \sum_{x \in \mathcal{B}^n} (-1)^{f(x)}$$

This means that wheter f is constant or balanced we will obtain a precise result:

- Constant:
$$\frac{1}{2^n} \sum_{x \in \mathcal{B}^n} (-1)^{f(x)} = \pm 1$$
- Balanced:
$$\frac{1}{2^n} \sum_{x \in \mathcal{B}^n} (-1)^{f(x)} = 0$$

Finally, with only one evaluation of f we know if it's constant or balanced.

4

Mathematical foundations of quantum

4.1 Vector spaces and Linear maps

Groups

A group is a set G with a multiplication operation that satisfies:

1. Associativity: $a(bc) = (ab)c$
2. Unit element: $\exists 1 \in G | \forall a \in G \; a1 = 1a = a$
3. Inverse: $\forall a \in G \; \exists a^{-1} | aa^{-1} = a^{-1}a = 1$

If the multiplication is commutative the group is called **abelian** group.

Vector spaces

A vector space is a set V with a vector sum and a scalar multiplication that is commutative, making V an abelian group.
The vector sum satisfies:

1. Associativity: $u + (v + w) = (u + w) + v$
2. Null vector: $\exists \overrightarrow{0} | \forall v \in V \; \overrightarrow{0} + v = v + \overrightarrow{0}$
3. Inverse: $\forall v \in V \; \exists -v | v + (-v) = \overrightarrow{0}$
4. Commutative

The scalar multiplication satisfies:

1. $\alpha(u + v) = \alpha u + \alpha v$
2. $(\alpha + \beta)v = \alpha v + \beta v$
3. $\alpha(\beta v) = (\alpha\beta)v$
4. $1v = v$
5. $0v = \overrightarrow{0}$

Linear map

A linear map between vector spaces V and W is a function $L : V \to W$ that satisfies:

$$L(\alpha u + \beta v) = \alpha L(u) + \beta L(v)$$

Given a vector space V, its **dual** V^* is the set of all linear maps $L : V \to \mathbb{C}$.

Inside a vector space we can define a **linear subspace** $W \subset V$ if vector sum and scalar multiplication are closed in W.

Dimension of a vector space

A set of vectors $v_1, \ldots, v_n \in V$ is linearly dependent if:

$$\exists(\alpha_1, \ldots, \alpha_n) \neq (0, \ldots, 0) | \sum_{i=1}^{n} \alpha_i v_i = 0$$

Otherwise the vectors are **linearly independent**.
A vector space is n-dimensional if it has a set of maximum n linearly independent vectors. This set of vectors is called a **basis set** for V.

Given a subspace $S \subset V$, the linear span of S, called $[S]$ is the set of all linear combinations of the vectors in S.

$$[S] = \left\{ \sum_{i=1}^{n} \alpha_i s_i \middle| s_i \in S, \alpha_i \in \mathbb{C}, n \in \mathbb{N} \right\}$$

Given the basis set $S = \{e_1, \ldots, e_n\}$ of a vector space S we have that $[S] = V$, so:

$$\forall v \in V \ \exists \alpha_1, \ldots, \alpha_n | v = \sum_{i=1}^{n} \alpha_i e_i$$

The coefficients α_i are unique for a basis set.

4.2 Scalar inner product

Scalar product

A scalar product over a vector space V is a function $\langle v, w\rangle : V \times V \to \mathbb{C}$ that maps $v, w \in V$ to a complex $\langle v, w\rangle$. The product satisfies:

1. $\langle u, \alpha v + \beta w\rangle = \alpha\langle u, v\rangle + \beta\langle u, w\rangle$
2. $\langle u, v\rangle = \overline{\langle v, u\rangle}$
3. $\langle u, v\rangle \geq 0$ and $\langle u, u\rangle = 0 \iff u = 0$

Given two vectors $u = (u_1, \ldots, u_n), v = (v_1, \ldots, v_n) \in \mathbb{C}^n$ the scalar product is defined as:

$$\langle u, v\rangle = \sum_{i=1}^{n} \overline{u_i} v_i$$

This can also be written as a matrix product:

$$\langle u, v\rangle = \begin{pmatrix} \overline{u_1} & \ldots & \overline{u_n} \end{pmatrix} \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$$

We can also derive that:

$$\langle \alpha u + \beta v, w\rangle = \overline{\alpha}\langle u, w\rangle + \overline{\beta}\langle v, w\rangle$$

The scalar prtoduct is often written with the Dirac notation $\langle u|v\rangle$, where $\langle u|$ is called "bra" and $|v\rangle$ is called "ket". While $|v\rangle$ represent the vector v, $\langle u|$ represent a map $V \to \mathbb{C}$.

Schwarz inequality

For any vectors $u, v \in V$ that are linearly dependent, we have that:

$$|\langle u|v\rangle| \leq \sqrt{\langle u|u\rangle \langle v|v\rangle}$$

The norm of a vector u is $||u|| = \sqrt{\langle u|u\rangle}$.

Triangle inequality

For any vectors $u, v \in V$, we have that:

$$||u + v|| \leq ||u|| + ||v||$$

Orthogonal vectors

Two vectors $u, v \in V$ are **orthogonal** if:

$$\langle u|v\rangle = 0$$

A set of vectors $\{u_1, \ldots, u_m\}$ is **orthonormal** if:

$$\forall i, j \in \{1, m\} \;\; \langle u|v\rangle = \delta_{ij}$$

where δ_{ij} is called the Kronecker delta and is defined as:

$$\delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

Given an orthonormal basis $\{e_1, \ldots, e_n\}$ for an n-dimensional vector space V, we can write:

$$\forall v \in V \;\; \exists \alpha_1, \ldots, \alpha_n | v = \sum_{i=1}^{n} \alpha_i e_i$$

We can compute the values α_i as follows:

$$\langle e_j|u\rangle = \left\langle e_j \middle| \sum_{i=1}^{n} \alpha_i e_i \right\rangle = \sum_{i=1}^{n} \alpha_i \langle e_j|e_i\rangle = \sum_{i=1}^{n} \alpha_i \delta_{ij} = \alpha_j$$

In Dirac notation we can write:

$$|u\rangle = \sum_{i=1}^{n} |e_i\rangle \langle e_i|u\rangle \implies \langle v|u\rangle = \sum_{i=1}^{n} \langle v|e_i\rangle \langle e_i|u\rangle$$

Given the above left equation we must have that:

$$\sum_{i=1}^{n} |e_i\rangle \langle e_i| = I$$

4.3 Hilber spaces

Convergence of vector sequence

A vector sequence $v_m \in V$ **converges strongly** to $v \in V$, written $v_m \to v$ if:

$$\lim_{m \to +\infty} ||v - v_m|| = 0$$

If $v \to v_m$ then:

- Weak convergence: $\lim_{m\to+\infty} ||v_m|| = ||v||$
- Scalar product continous: $\langle u|v\rangle = \lim_{m\to+\infty} \langle u|v_m\rangle$

Cauchy sequence

A vector sequence $v_i \in V$ is a **Cauchy sequence** if:

$$\forall \epsilon > 0 \ \exists n_\epsilon | \forall n, m > n_\epsilon \ ||v_n - v_m|| < \epsilon$$

Hilbert space

A Hilbert space $\mathcal{H}$ is a complete scalar product space, so every Cauchy sequence converges strongly to an element in the space.
Finite-dimensional vector spaces are always complete.

Hilbert spaces also exists for infinite dimensional spaces but for quantum computing only the finite ones are needed.

Topologically closed subspaces

A linear subspace W of a Hilbert space $\mathcal{H}$ is **topologically closed** if any sequence of vectors in W converges in W.

Linear operator

Given an Hilber space $\mathcal{H}$ a linear **operator** is a linear function $A : \mathcal{H} \to \mathcal{H}$ that has two properties:

- Operator sum: $\forall v \in \mathcal{H} \ (A + B)v = Av + Bv$
- Operator product: $\forall v \in \mathcal{H} \ (AB)v = A(Bv)$

For operators A, B and $\lambda \in \mathbb{C}$, we have:

1. $(AB)^\dagger = B^\dagger A^\dagger$
2. $(\lambda A)^\dagger = \overline{\lambda} A^\dagger$
3. $(A^\dagger)^\dagger = A$

The adjoint of an operator A is the operator $A^\dagger$ defined as:

$$\forall u, v \in \mathcal{H} \ \left\langle u \middle| A^\dagger v \right\rangle = \langle Au | v \rangle$$

An operator is called **self-adjoint** (or Hermitian) if $A = A^\dagger$. If an operator is self-adjoint we have:

$$\langle u | Av \rangle = \langle Au | v \rangle = \overline{\langle v | Au \rangle}$$

Any linear operator A on a scalar product vector space with orthonormal basis can be represented in matrix form where:

$$A_{ij} = \langle e_i | A e_j \rangle \qquad A^\dagger_{ij} = \overline{A_{ij}}$$

Unitary operators

We can define an operator as unitary in two ways:

- An operator U is unitary if satisfies:

$$UU^\dagger = U^\dagger U = I$$

- An operator U is unitary if satisfies:

  1. U is surjective
  2. $\forall x \in \mathcal{H}\ ||Ux|| = ||x||$

An operator is **normal** if:

$$A^\dagger A = AA^\dagger$$

Both self-adjoint and unitary operators are normal.

Eigenvalues of operators

An eigenvalue λ of an operator A is d-fold generative if there are d linearly independent eigenvectors associated with λ.
The eigenvectors of a given eigenvalue form a linear subspace, called **eigenspace**.

In Dirac notation if λ is a non-degenerate eigrnvalue of A, we write that the only eigenvector associated to λ as $|\lambda\rangle$ and we have that:

$$A\,|\lambda\rangle = \lambda\,|\lambda\rangle$$

From the equation above we can derive:

$$\langle\lambda|\,A^\dagger = \overline{\lambda}\,\langle\lambda|$$

Properties of eigenvalues of operators

Eigenvalues for self-adjoint and unitary operators have some properties:

1. Eigenvalues of self-adjoint operators are real numbers.
2. Eigenvalues of unitary operators are complex numbers of modulus 1.
3. Eigenvectors of self-adjoint and unitary matrices associated to different eigenvectors are orthogonal.

4.3.1 Spectral Theory

Spectral theorem

The set of all eigenvectors u_{ij} of a normal operator is an orthogonal basis for $\mathcal{H}$, so:

$$\forall v \in \mathcal{H}\ v = \sum_{i=1}^{m}\sum_{j=1}^{d_i} \alpha_{ij} u_{ij} \qquad \alpha_{ij} = \langle u_{ij}|v\rangle$$

where m is the number of eigenvalues and d_i is the degenerate of the i-th eigenvalue.

We have that the dimension of $\mathcal{H}$ is:

$$\dim \mathcal{H} = \sum_{i=1}^{m} d_i$$

In Dirac notation we can write:

$$|v\rangle = \sum_{i=1}^{m}\sum_{j=1}^{d_i} \langle u_{ij}|v\rangle\, |\lambda_{ij}\rangle$$

where λ_{ij} is the j-th eigenvector associated to the i-th eigenvalue.

4.4 Projectors

Orthogonal subspaces

Two linear subspaces $V, W \subset \mathcal{H}$ are orthogonal if every vector of V is orthogonal to every vector of W.
The orthogonal complement of V is:

$$V^{\perp} = \{u \in \mathcal{H} | \forall v \in V\ \ \langle u|v\rangle = 0\}$$

For finite-dimensional Hilbert spaces $(V^{\perp})^{\perp} = V$.

Given a closed subspace $W \subset \mathcal{H}$ we want to write any vector $v \in \mathcal{H}$ as a sum $v_W + v_{W^{\perp}}$ where $v_W \in W, v_{W^{\perp}} \in W^{\perp}$.

Projectors

Let f_i be an orthogonal basis for W, we define:

$$\forall v \in \mathcal{H}\ v_w = \sum_i \langle f_i|v\rangle\, f_i \qquad v_{W^{\perp}} = v - v_W$$

The sum is called the **projector operator** on W, a map $P_W : \mathcal{H} \to W$ defined as $P_W(v) = v_W$. The projector on the orthogonal complement of W is $P_{W^{\perp}} = I - P_W$.

For a projector we have that:

$$P_W(v) = \begin{cases} v & v \in W \\ 0 & v \in W^{\perp} \end{cases}$$

So P_W has only two eigenvalues: 0 and 1.
Two projectors P, Q are orthogonal if and only if:

$$PQ = QP = 0$$

If $P \perp Q$, then $P + Q$ is a projector.
Given a family of projectors P_i such that $P_i P_j = \delta_{ij}$, then:

$$\sum_i P_i = I$$

Function of an operator

Given a self-adjoint operator A and a function $f : \mathbb{R} \to \mathbb{C}$, the operator $f(A)$ is defined by:

$$f(A) = \sum_{j=1}^{m} f(\lambda_j) P_j$$

where λ_j is the j-th eigenvalue of A and P_j is the projector on the eigenspace of λ_j.

For a self-adjoint operator A we have that e^{iA} is unitary.

4.5 Tensor product

Tensor product

Given two n-dimensional vectors u, v, the tensor product is a n^2-dimensional vector w defined by:

$$w_i = u_{i \text{ div } n} \cdot v_{i \mod n}$$

where i div n is the integer division.

The scalar product of tensor vectors is defined by:

$$\langle u \otimes v | w \otimes z \rangle = \langle u | v \rangle \langle v | z \rangle$$

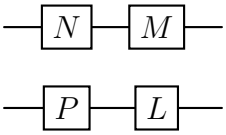
The tensor product of operators A, B is defined by:

$$(A \otimes B) u \otimes v = Au \otimes Bv$$

For suitably sized matrices or operators L, M, N, P we have that:

$$(M \cdot N) \otimes (L \cdot P) = (M \otimes L) \cdot (N \otimes P)$$

This equation can be pictured as a quantum circuit:



4.6 Rules of Quantum Mechanics

There are 4 rules of Quantum Mechanics:

1. The state space of Quantum Mechanics is a Hilbert space. So, any quantum system "lives" in a Hilbert space.

2. Observables are represented mathematically by self-adjoint operators on $\mathcal{H}$.

3. The third rule can be written in two ways:

   - Given an observable A and a state $v \in \mathcal{H}$ the expected result of measuring A is $\langle v|Av\rangle$.
   - Given an observable A and a state $v \in \mathcal{H}$ the only possible results of measuring A are one of its eigenvalues. Also, the probability of measuring a specific eigenvalue λ in state v is:
     $$P(A = \lambda; v) = \langle v|P_\lambda v\rangle$$
     where P_λ is the projector on the eigenspace of λ.

4. A closed system evolves through time according to the Schrodinger equation:
   $$i\hbar\frac{dv(t)}{dt} = Hv(t)$$
   where $\hbar$ is the planck constant (a small number), $v(t)$ is a vector depending on time and H is a self-adjoint operation describing the total energy of the system (called Hamiltonian).

4.6.1 Quantum evolution is Unitary

Schrodinger equation is a linear differential equation:

$$i\hbar\frac{dv(t)}{dt} = Hv(t)$$

For any $t \geq t_0 \geq 0$, the solution $v(t)$ is:

$$v(t) = e^{-\frac{i}{\hbar}H(t-t_0)}v(t_0)$$

By choosing the initial value $v(t_0)$ we can obtain a solution for any time point.
The term before $v(t_0)$ is a function of the operator H, that can be written as:

$$e^{-\frac{i}{\hbar}H(t-t_0)} = \sum_{j=1}^{m}\left|e^{-\frac{i}{\hbar}\lambda_j(t-t_0)}\right|P_j$$

where λ_j is the j-th eigenvalue and P_j the projector on the eigenspace.
This means that quantum evolution is a unitary operator.

5

Search problem

Suppose we have an array of lenght $N = 2^n$, with $M \in [1, N]$ solution elements and we want to find the index of a solution element.
Using a classical random algorithm we would need $O\left(\frac{N}{M}\right)$ accesses to the array in the worst case.

5.1 Grover algorithm

The Grover algorithm returns a correct index with high probability ($> 50\%$) with only $O\left(\sqrt{\frac{N}{M}}\right)$.
The array has size $N = 2^n$ so the indexes of the array can be encoded using n bits. We use a boolean function to encode the solutions:

$$f(i) = \begin{cases} 0 & A[i] \text{ is not a solution} \\ 1 & A[i] \text{ is a solution} \end{cases}$$

For the function f we build the unitary U_f:

$$U_f : |x \otimes y\rangle \to |x \otimes (y \otimes f(x))\rangle$$

We also know that:

$$U_f |x0\rangle = |xf(x)\rangle \qquad U_f |x1\rangle = |x\neg f(x)\rangle$$

Using $y = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ we have that:

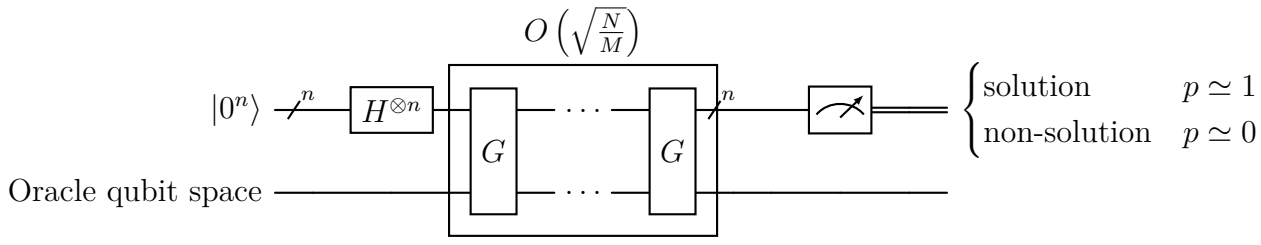
$$U_f |x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = |x\rangle \otimes \frac{1}{\sqrt{2}}(|f(x)\rangle - |\neg f(x)\rangle)$$

$$= \begin{cases} |x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) & f(x) = 0 \\ |x\rangle \otimes \frac{1}{\sqrt{2}}(-|0\rangle + |1\rangle) & f(x) = 1 \end{cases} = (-1)^{f(x)} |x\rangle \otimes \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

The right qubit is unaltered so we can drop it and obtain the **oracle**:

$$O_f |x\rangle = (-1)^{f(x)} |x\rangle$$

This oracle flips the amplitude of all solution elements.
The Grover algorithm is a oracle (or black-box) quantum algorithm, described by this circuit:



The state of n qubits is represented as:

$$\sum_{x \in \mathcal{B}^n} \alpha_x |x\rangle$$

The gate G is described as:

$$G = W \cdot O_f \qquad W|x\rangle = (-\alpha_x + 2\langle a \rangle)|x\rangle$$

where $\langle a \rangle$ is the mean amplitude:

$$\langle a \rangle = \frac{1}{N} \sum_{x \in \mathcal{B}^n} \alpha_x$$

We define the set of solution indeces S and two vectors:

$$|a\rangle = \frac{1}{\sqrt{N-M}} \sum_{x \in \bar{S}} |x\rangle \qquad |b\rangle = \frac{1}{\sqrt{M}} \sum_{x \in S} |x\rangle$$

This two vectors describe the equally weighted superposition of all solution and non-solution indeces respectively. This means that the equally weighted superposition created by the $H^{\otimes n}$ gate can be written as:

$$|\psi\rangle = H^{\otimes n}|0^n\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \mathcal{B}^n} |x\rangle = \sqrt{\frac{N-M}{N}}|a\rangle + \sqrt{\frac{M}{N}}|b\rangle$$

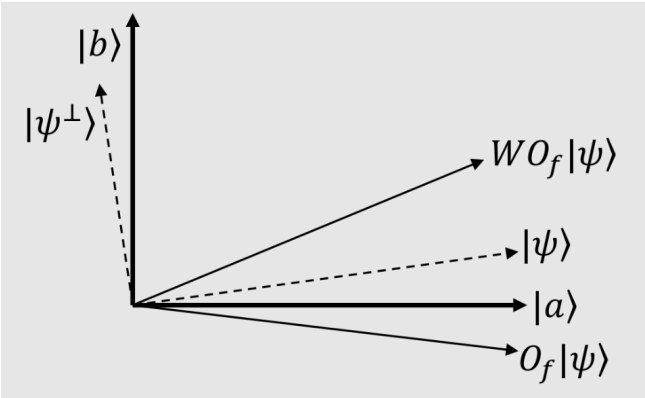
Applying G on this qubit means applying O_f first:

$$|\psi\rangle = \sqrt{\frac{N-M}{N}}|a\rangle + \sqrt{\frac{M}{N}}|b\rangle \xrightarrow{O_f} \sqrt{\frac{N-M}{N}}|a\rangle - \sqrt{\frac{M}{N}}|b\rangle$$

The operator O_f performs a reflection of $|\psi\rangle$ with respect to $|a\rangle$.
We can rewrite W as:

$$W = 2P_\psi - I = 2P_\psi - (P_\psi + P_{\psi^\perp}) = P_\psi - P_{\psi^\perp}$$

So, W performs a reflection with respect to the original $|\psi\rangle$.
Graphically we can describe G as a rotation on a plane defined by $|a\rangle$ and $|b\rangle$.



To better explain G let $\frac{\theta}{2}$ be the angle between $|a\rangle$ and $|\psi\rangle$. Given the definition of $|\psi\rangle$ we can find an angle θ such that:

$$\cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{N-M}{N}} \qquad \sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{M}{N}}$$

Since G is a rotation by θ we have that:

$$G|\psi\rangle = WO_f|\psi\rangle = \cos\left(\frac{3\theta}{2}\right)|a\rangle + \sin\left(\frac{3\theta}{2}\right)|b\rangle$$

The application of k gates results in:

$$G^k|\psi\rangle = \cos\left(\frac{(2k+1)\theta}{2}\right)|a\rangle + \sin\left(\frac{(2k+1)\theta}{2}\right)|b\rangle$$

The angle between $|\psi\rangle$ and $|b\rangle$ that we want to rotate is:

$$\cos^{-1}\left(\sqrt{\frac{M}{N}}\right)$$

Each application of G rotates by θ so the number of time we need to apply G is:

$$k = \left\lfloor \frac{\cos^{-1}\left(\sqrt{\frac{M}{N}}\right)}{\theta} \right\rfloor$$

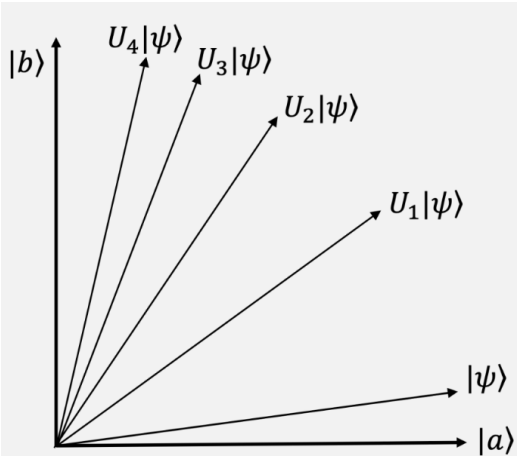
This gets the qubit within an angle of $\frac{\pi}{4}$ of $|b\rangle$ so, when measuring there is a probability to get a solution of at least 50%.
If $M \leq \frac{N}{2}$, it means that $\theta \leq \frac{\pi}{2}$. Since $\cos^{-1} \leq \frac{\pi}{2}$ we have that $k \leq \left\lceil \frac{\pi}{2\theta} \right\rceil$. We also have that $\sqrt{\frac{M}{N}} = \sin\left(\frac{\theta}{2}\right) \leq \frac{\theta}{2}$ that implies $\frac{1}{\theta} \leq \frac{1}{2}\sqrt{\frac{N}{M}}$ and thus:

$$k = \left\lfloor \frac{\cos^{-1}\left(\sqrt{\frac{M}{N}}\right)}{\theta} \right\rfloor \leq \left\lceil \frac{\pi}{2\theta} \right\rceil \leq \left\lceil \frac{\pi}{4}\sqrt{\frac{N}{M}} \right\rceil$$

To remove the constraint $M \leq \frac{N}{2}$ we can double the size of the array adding fake non-solution values to reinsert the constraint, this requires only 1 more qubit.
If we don't know the solution is in the array, so M could also be 0, we can add a fake solution to the array. Since all the solution are measured with the same probability if we always get the fake solution is likely that there wasn't solution in the starting array.

5.2 Fixed-Point quantum search

Grover algorithm require to know the number of solution elements in the array to know when to stop applying the G gate.
There is another algorithm, called **fixed-point quantum search**, that monotonically moves the starting qubit $|\psi\rangle$ toward the target.



Define the phase shift operators:

$$R_s = I - (1 - e^{i\theta}) \overbrace{|s\rangle \langle s|}^{P_s}$$
$$R_t = I - (1 - e^{i\theta}) \underbrace{|t\rangle \langle t|}_{P_t}$$

where $\theta = \frac{\pi}{3}$, $|s\rangle$ is the starting state and $|t\rangle$ is the target state.
Let U be any unitary operator such that, for a small $\epsilon > 0$:

$$|U_{ts}|^2 = |\langle t|Us\rangle|^2 = 1 - \epsilon$$

Define the unitary operator:

$$G = UR_sU^\dagger R_tU$$

Probability of measuring $|t\rangle$

Given the operator G, the probability of measuring $|t\rangle$ when starting from $|s\rangle$ is:

$$\left|\langle t| UR_sU^\dagger R_tU |s\rangle\right|^2 = 1 - \epsilon^3$$

Proof:
It can be shown that:

$$UR_sU^\dagger R_tU |s\rangle = U |s\rangle \left[e^{i\theta} + |U_{ts}|^2(e^{i\theta} - 1)^2\right] + |t\rangle U_{ts}(e^{i\theta} - 1)$$

To know how close the state above is to $|t\rangle$ we need to compute the squared norm of:

$$\begin{aligned}
&\left|\langle t| \left[U |s\rangle \left(e^{i\theta} + |U_{ts}|^2(e^{i\theta} - 1)^2\right) + |t\rangle U_{ts}(e^{i\theta} - 1)\right]\right|^2 \\
&= \left|\langle t| U |s\rangle \left(e^{i\theta} + |U_{ts}|^2(e^{i\theta} - 1)^2\right) + \langle t|t\rangle U_{ts}(e^{i\theta} - 1)\right|^2 \\
&= \left|U_{ts} \left(e^{i\theta} + |U_{ts}|^2(e^{i\theta} - 1)^2\right) + U_{ts}(e^{i\theta} - 1)\right|^2 \\
&= \left|U_{ts} \left[2e^{i\theta} - 1 + |U_{ts}|^2(e^{i\theta} - 1)^2\right]\right|^2 = |U_{ts}|^2 \cdot \left|2e^{i\theta} - 1 + |U_{ts}|^2(e^{i\theta} - 1)^2\right|
\end{aligned}$$

By putting $\theta = \frac{\pi}{3}$ and $|U_{ts}|^2 = 1 - \epsilon$ we get:

$$|U_{ts}|^2 \cdot \left|2e^{i\theta} - 1 + |U_{ts}|^2(e^{i\theta} - 1)^2\right| = (1 - \epsilon) \left|2e^{i\frac{\pi}{3}} - 1 + (1 - \epsilon)(e^{i\frac{\pi}{3}} - 1)^2\right|^2$$

With some transformation this is equal to $1 - \epsilon^3$ so, the probability of measuring $|t\rangle$ after appling G is incresed from $1 - \epsilon$ to $1 - \epsilon^3$.
Given U satisfying $|U_{ts}|^2 = 1 - \epsilon$ we define the recursive sequence of operators:

$$\begin{cases} U_0 = U & m = 0 \\ U_m = U_{m-1} R_s U_{m-1}^\dagger R_t U_{m-1} & m \geq 1 \end{cases}$$

It can be shown that the probability of measuring $|t\rangle$ starting from $|s\rangle$ after applying U_m is:

$$|\langle t|U_m s\rangle|^2 = 1 - \epsilon^{2q_m+1}$$

where q_m is the number of queries to the oracle.
This is similar to the classical probabilistic algorithm where the failure probability drops to ϵ^{q+1} after q oracle queries.

6

Quantum Fourier Transform

6.1 Polynomials multiplication

Given two polynomials of degree $n-1$ defined as:

$$p(x) = \sum_{i=0}^{n-1} a_i x^i \qquad q(x) = \sum_{j=0}^{n-1} b_j x^j$$

Their product is computed as:

$$p(x)q(x) = \left(\sum_{i=0}^{n-1} a_i x^i\right)\left(\sum_{j=0}^{n-1} b_j x^j\right) = \sum_{i,j=0}^{n-1} a_i x^{i+j} b_j$$

By summing the monomials of the same degree we obtain:

$$p(x)q(x) = \sum_{k=0}^{2n-2} x^k \left(\sum_{j=0}^{k} a_j b_{k-j}\right) \qquad a_j = b_j = 0 \text{ if } j < 0 \text{ or } j \geq n$$

This can also be written as:

$$p(x)q(x) = \sum_{k=0}^{2n-1} x^k c_k$$

Convolution

Give two vectors a, b of size n the convolution c is the vector $c = a \circledast b$ with size $2n$ defined by:

$$c_k = \sum_{j=0}^{k} a_j b_{k-j} \qquad a_j = b_j = 0 \text{ if } j < 0 \text{ or } j \geq n$$

for $k \in [0, 2n-1]$.

6.2 Discrete Fourier Transform

The **Discrete Fourier Transform (DFT)** is defined over an arbitrary commutative ring $(R, \cdot, +, 0, 1)$.

n-th root of unity

A $w \in R$ is a principal n-th root of unity if:

1. $w \neq 1$
2. $w^n = 1$
3. $\sum_{j=0}^{n-1} w^{j \cdot p} = 0 \ \forall p \in [1, n-1]$

The powers $w^0, \ldots, w^{n-1}$ are the n-th root of unity.

Usually, we use $w = e^{\frac{2\pi i}{n}}$ for $n \in \mathbb{N}$.
Let R be a commutative ring and $w \in R$ a principal n-th root of unity. We define a $n \times n$ matrix A:

$$A_{ij} = w^{i \cdot j} \quad \text{for } i, j \in [0, n-1]$$

Discrete Fourier Transform

Let $a \in R^n$ be a vector in a commutative ring R. The Discrete Fourier Transform (DFT) of a is the vector $F(a)$ defined as:

$$F(a) = A \cdot a$$

The i-th element of $F(a)$ is:

$$F(a)_i = \sum_{k=0}^{n-1} A_{ik} a_k = \sum_{k=0}^{n-1} a_k w^{i \cdot k}$$

The DFT of a vector is the conversion of a polynomial from the coefficient representation to its value representation in the points $w_0, \ldots, w^{n-1}$.

The inverse DFT is the matrix A_{ij}^{-1} given by:

$$A_{ij}^{-1} = \frac{1}{n} w^{-i \cdot j} \quad \text{for } i, j \in [0, n-1]$$

Convolution theorem

Let $a = (a_0, \ldots, a_{n-1}, 0, \ldots, 0) \in R^{2n}$ and $b = (b_0, \ldots, b_{n-1}, 0, \ldots, 0) \in R^{2n}$ be two vectors of size $2n$, we have that:

$$a \circledast b = F^{-1}(F(a) \cdot F(b))$$

6.3 Quantum Fourier Transform Definition

Quantum Fourier Transform

The Quantum Fourier Transform (QFT) maps each basis state $|0\rangle, |1\rangle, \ldots, |N-1\rangle$ as follows:

$$QFT\,|x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i x k}{N}} |k\rangle$$

The QFT matrix can be written as:

$$QFT = \sum_{j=0}^{N-1} \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{\frac{2\pi i j k}{N}} |k\rangle \right) \langle j|$$

QFT unitary

The QFT is a unitary operator and is adjoint is:

$$QFT^{\dagger} = \sum_{j=0}^{N-1} |j\rangle \left(\frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{-\frac{2\pi i j k}{N}} \langle k| \right)$$

Proof:
To prove that QFT is unitary we need to show that:

$$QFTQFT^{\dagger} = QFT^{\dagger}QFT = I$$

We will prove only the second product:

$$\begin{aligned} QFT^{\dagger}QFT &= \sum_{j=0}^{N-1} |j\rangle \left(\frac{1}{\sqrt{N}} \sum_{r=0}^{N-1} e^{-2\pi i j r/N} \langle r| \right) \sum_{k=0}^{N-1} \left(\frac{1}{\sqrt{N}} \sum_{s=0} N - 1 e^{2\pi i s k/N} |s\rangle \right) \langle k| \\ &= \frac{1}{N} \sum_{j,k=0}^{N-1} |j\rangle \left(\sum_{r=0}^{N-1} e^{-2\pi i j r/N} \langle r| \right) \left(\frac{1}{\sqrt{N}} \sum_{s=0} N - 1 e^{2\pi i s k/N} |s\rangle \right) \langle k| \\ &= \frac{1}{N} \sum_{j,k=0}^{N-1} |j\rangle \left(\sum_{r,s=0}^{N-1} e^{2\pi i(-jr+ks)} \langle r|s\rangle \right) \langle k| \end{aligned}$$

Recalling that $\langle r|s\rangle = \delta_{rs}$, we can consider only the indeces in the sum where $r = s$, so:

$$\frac{1}{N} \sum_{j,k=0}^{N-1} |j\rangle \left(\sum_{r,s=0}^{N-1} e^{2\pi i(-jr+ks)/N} \langle r|s\rangle \right) \langle k| = \frac{1}{N} \sum_{j,k=0}^{N-1} |j\rangle \left(\sum_{r=0}^{N-1} e^{2\pi i r(k-j)/N} \right) \langle k|$$

If $j = k$ then the exponent of e will be 0, so we can divide the sum considering this case:

$$\begin{aligned} &\frac{1}{N} \sum_{j,k=0;j=k}^{N-1} |j\rangle \left(\sum_{r=0}^{N-1} 1 \right) \langle k| + \frac{1}{N} \sum_{j,k=0;j\neq k}^{N-1} |j\rangle \left(\sum_{r=0}^{N-1} e^{2\pi i r(k-j)/N} \right) \langle k| \\ &= \sum_{j=0}^{N-1} |j\rangle \langle j| + \frac{1}{N} \sum_{j,k=0;j\neq k}^{N-1} |j\rangle \left(\sum_{r=0}^{N-1} \left(e^{2\pi i(k-j)/N} \right)^r \right) \langle k| \end{aligned}$$

We know that for any value $\rho \neq 1$ we can write the sum:

$$\sum_{i=0}^{N-1} \rho^i = \frac{1-\rho^N}{1-\rho}$$

Since the cases where the value inside the sum is 1 were already splitted before we can use this formula:

$$\begin{aligned}
&\sum_{j=0}^{N-1} |j\rangle\langle j| + \frac{1}{N} \sum_{j,k=0;j\neq k}^{N-1} |j\rangle \left(\sum_{r=0}^{N-1} \left(e^{2\pi i(k-j)/N} \right)^r \right) \langle k| \\
&= I + \frac{1}{N} \sum_{j,k=0;j\neq k}^{N-1} |j\rangle \left(\frac{1-(e^{2\pi i(k-j)/N})^N}{1-e^{2\pi i(k-j)/N}} \right) \langle k| \\
&= I + \frac{1}{N} \sum_{j,k=0;j\neq k}^{N-1} |j\rangle \left(\frac{1-e^{2\pi i(k-j)}}{1-e^{2\pi i(k-j)/N}} \right) \langle k|
\end{aligned}$$

For every value $j \neq k$ the numerator is always $1-1=0$ so the sum disappears leaving only the identity.
An equivalent QFT definition for $N = 2^n$, so n qubit is:

$$QFT\,|j_1 \dots j_n\rangle = \frac{(|0\rangle + e^{2\pi i(0.j_n)}\,|1\rangle) \otimes (|0\rangle + e^{2\pi i(0.j_{n-1}j_n)}\,|1\rangle) \otimes \cdots \otimes (|0\rangle + e^{2\pi i(0.j_1 \dots j_n)}\,|1\rangle)}{\sqrt{2^n}}$$

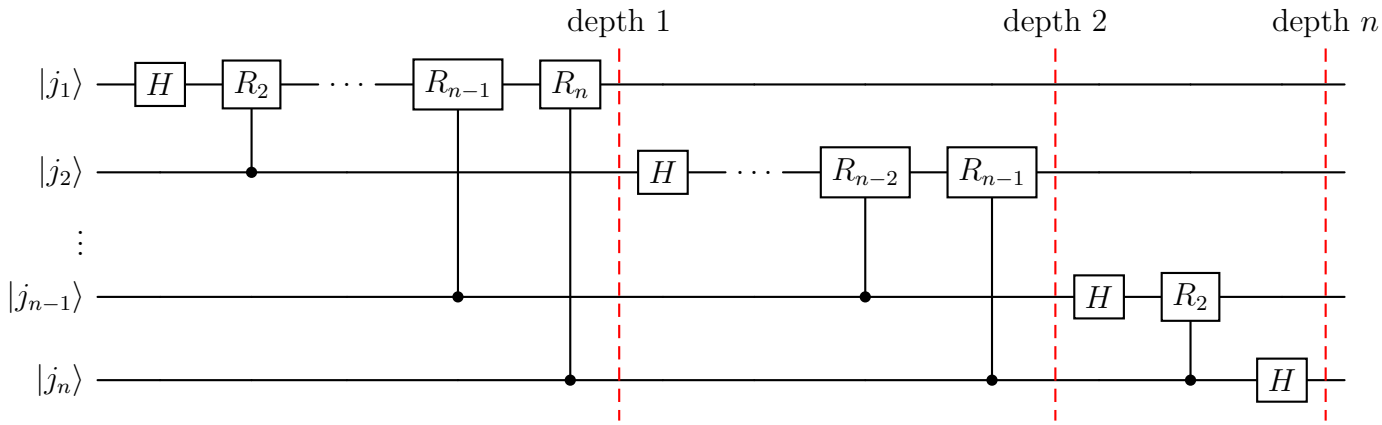
where $j_1, \dots, j_n$ are bits and the binary fraction $0.j_l \dots j_m$ is defined as:

$$0.j_l \dots j_m = \frac{j_l}{2} + \cdots + \frac{j_m}{2^{m-l+1}}$$

6.4 QFT circuit

The QFT quantum circuit uses only Hadamard and controlled-R_k gates defined on:

$$R_k = \begin{pmatrix} 1 & 0 \\ 0 & e^{2\pi i/2^k} \end{pmatrix}$$



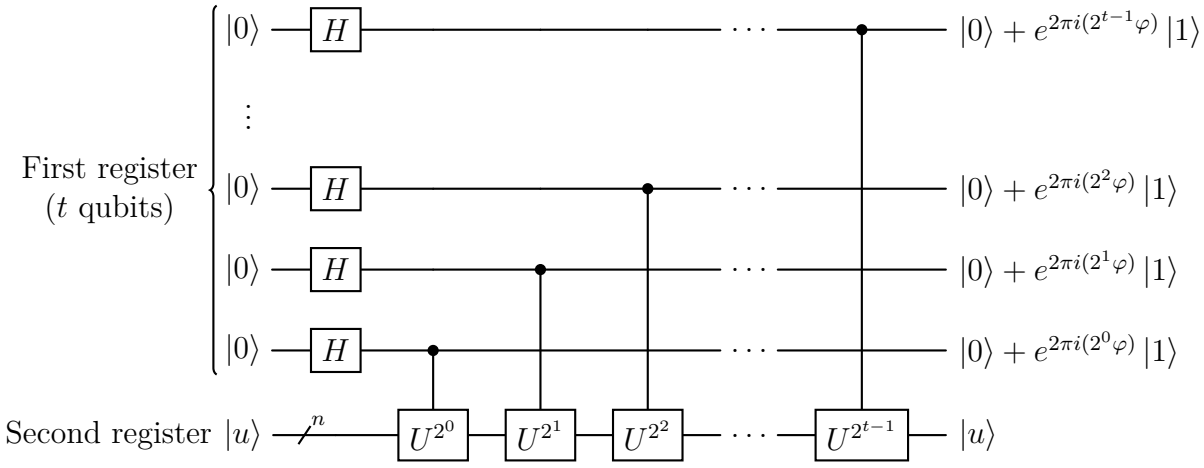
The state at different depth are:

- Depth 1:
$$\frac{1}{2^{1/2}}(|0\rangle + e^{2\pi i(0.j_1 \dots j_n)}|1\rangle)\,|j_2 \dots j_n\rangle$$
- Depth 2:
$$\frac{1}{2^{2/2}}(|0\rangle + e^{2\pi i(0.j_1 \dots j_n)}|1\rangle) \otimes (|0\rangle + e^{2\pi i(0.j_2 \dots j_n)}|1\rangle)\,|j_3 \dots j_n\rangle$$
- Depth n:
$$\frac{1}{2^{n/2}}(|0\rangle + e^{2\pi i(0.j_1 \dots j_n)}|1\rangle) \otimes (|0\rangle + e^{2\pi i(0.j_2 \dots j_n)}|1\rangle) \otimes \dots \otimes (|0\rangle + e^{2\pi i(0.j_n)}|1\rangle)$$

The depth n has all the correct term but in reverse order, we need to swap them unitarly.
The circuit has $O(n^2)$ gates while the best classical circuit needs $O(n2^n)$ gates.

6.5 Quantum Phase Estimation algorithm

The eigenvalue of any unitary operator are compex number λ of modulus 1, so they can be written as:
$$\lambda = e^{2\pi i\varphi} \qquad \varphi \in [0,1]$$
Finding φ is a problem that can be solved with QFT.
Let u be the eigenvector associated to an eigenvalue $e^{2\pi i\varphi}$ of a unitary operator U, so:
$$U|u\rangle = \lambda|u\rangle = e^{2\pi i\varphi}|u\rangle$$



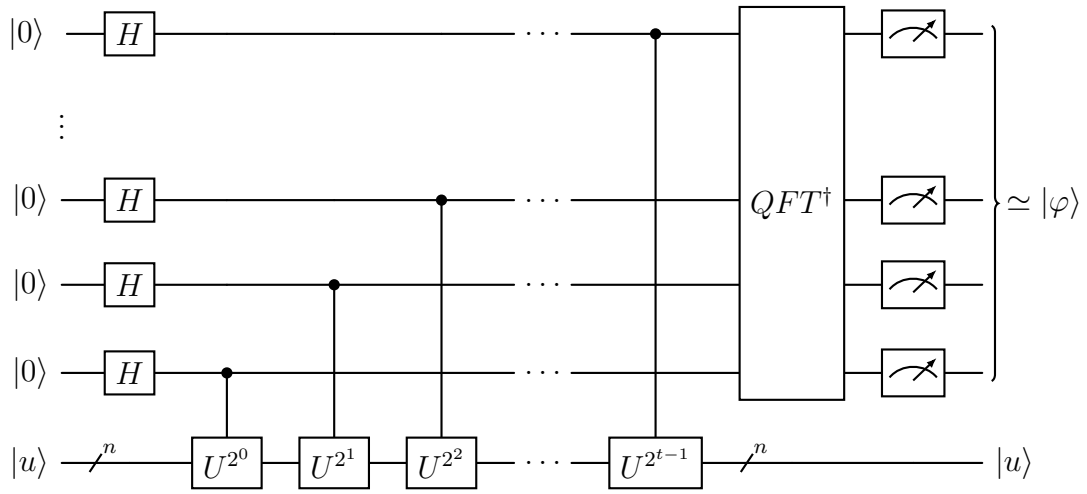
The gate U^{2^k} applies the U operator 2^k times.
The state of the firts t qubits at the end of the circuit is:
$$\frac{1}{2^{t/2}}(|0\rangle + e^{2\pi i(2^{t-1}\varphi)}|1\rangle) \otimes \dots \otimes (|0\rangle + e^{2\pi i(2^0\varphi)}|1\rangle) = \frac{1}{2^{t/2}}\sum_{k=0}^{2^{t-1}} e^{2\pi ik\varphi}|k\rangle$$

Suppose that we can write φ with exactly t bits:
$$\varphi = 0.\varphi_1 \dots \varphi_t$$

Then the state at the end of the QPE algorithm is:

$$\frac{1}{2^{t/2}}(|0\rangle + e^{2\pi i(0.\varphi_t)}|1\rangle) \otimes \cdots \otimes (|0\rangle + e^{2\pi i(0.\varphi_1 \ldots \varphi_t)}|1\rangle)$$

If we apply the $QFT^\dagger$ at the end of the algorithm we will measure the phase $|\varphi_1 \ldots \varphi_t\rangle$ with probability 1. The final quantum circuit to solve phase estimation is:



Approximation of φ

If φ cannot be expressed with exactly t bits we can estimate it with n bits of precision and success probability of at least $1 - \epsilon$ using r qubits. The number of qubits used is:

$$r = n + \left\lceil \log\left(2 + \frac{1}{2\epsilon}\right) \right\rceil$$

7

Integer Factoring

7.1 Order-finding problem

Order of a number

Let x, N two numbers $x < N$ that are coprime, so $\text{GCD}\, x, N = 1$. The **order** of x modulo N is the least integer r such that:

$$x^r = 1 \mod N$$

The **Order-finding problem** is given x and N to find r.
There are no classical algorithms that are polynomial in the input lenght $\log N$ (bits of N), but a quantum algorithm exists.

7.1.1 Modular multiplication operator

We define the unitary operator U_x that operates on $L = \lceil \log N \rceil$ bits:

$$U_x |y\rangle = \begin{cases} |xy \mod N\rangle & y \leq N \\ |y\rangle & y > N \end{cases}$$

Equivalence of modular multiplication

If x and N are coprime, we have that:

$$\forall y, z \in \mathbb{Z}\ xy = xz \mod N \iff y = z \mod N$$

Unitarity of modular multiplication

The operator U_x is unitary.

Proof:
We need to prove that:

$$U_x U_x^\dagger = U_x^\dagger U_x = I$$

where U_x and $U_x^\dagger$ are defined as:

$$U_x = \sum_{y \leq N} U_x |y\rangle \langle y| + \sum_{y > N} |y\rangle \langle y| \qquad U_x^\dagger = \sum_{y \leq N} |y\rangle (U_x |y\rangle)^\dagger + \sum_{y > N} |y\rangle \langle y|$$

We will prove only the second product:

$$U_x^\dagger U_x = \left(\sum_{y\leq N} |y\rangle (U_x |y\rangle)^\dagger + \sum_{y>N} |y\rangle\langle y|\right) + \left(\sum_{z\leq N} U_x |z\rangle\langle z| + \sum_{z>N} |z\rangle\langle z|\right)$$
$$= \left(\sum_{y\leq N} |y\rangle\langle U_x y| + \sum_{y>N} |y\rangle\langle y|\right) + \left(\sum_{z\leq N} |U_x z\rangle\langle z| + \sum_{z>N} |z\rangle\langle z|\right)$$
$$= \sum_{y,z\leq N} |y\rangle |U_x y\rangle\langle U_x z|\langle z| + \sum_{y\leq N;z>N} |y\rangle |U_x y\rangle\langle z|\langle z| + \sum_{y>N;<\leq N} |y\rangle |y\rangle\langle U_x z|\langle z| + \sum_{y,z>N} |y\rangle |y\rangle\langle z|\langle z|$$

In the second sum $z > N$ and $y \leq N$, it means that $U_x y = xy \mod N$ is lower that z, so $|U_x y\rangle\langle z| = 0$ since they are orthogonal. Using the same strategy $|y\rangle\langle U_x z| = 0$ since they are orthogonal and the two sums are deleted.

$$\sum_{y,z\leq N} |y\rangle |U_x y\rangle\langle U_x z|\langle z| + \overbrace{\sum_{y\leq N;z>N} |y\rangle |U_x y\rangle\langle z|\langle z|}^{0} + \overbrace{\sum_{y>N;<\leq N} |y\rangle |y\rangle\langle U_x z|\langle z|}^{0} + \sum_{y,z>N} |y\rangle |y\rangle\langle z|\langle z|$$
$$= \sum_{y,z\leq N} |y\rangle |U_x y\rangle\langle U_x z|\langle z| + \sum_{y,z>N} |y\rangle |y\rangle\langle z|\langle z|$$

In the first sum we could write $|Uxy\rangle\langle U_x z| = |xy \mod N\rangle\langle xz \mod N|$. Using the previous definition the two parts are equal only if $y = z$ and the result is 1, in all the other cases the result is 0. In the same ways $|y\rangle\langle z| = 1$ only if $y = z$ and 0 otherwise. So, from both sums we can delete all the terms where $y \neq z$.

$$\sum_{y,z\leq N} |y\rangle |U_x y\rangle\langle U_x z|\langle z| + \sum_{y,z>N} |y\rangle |y\rangle\langle z|\langle z|$$
$$= \sum_{y=z\leq N} |y\rangle\langle z| + \sum_{y,z>N} |y\rangle\langle z| = \sum_{y\leq N} |y\rangle\langle y| + \sum_{y>N} |y\rangle\langle y| = I$$

Eigenvector of modular multiplication

The eigenvectors of U_x can be written in the form:

$$\forall s \in [0, r-1] \quad |u_s\rangle = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi i s k/r} \left|x^k \mod N\right\rangle$$

where r is the order of x modulo N.

Proof:
We need to find an eigenvalue $\lambda \in \mathbb{C}$ such that $U_x |u_s\rangle = \lambda |u_s\rangle$.

$$U_x |u_s\rangle = U_x \left(\frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi isk/r} \left| x^k \mod N \right\rangle \right) = \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi isk/r} U_x \left| x^k \mod N \right\rangle$$
$$= \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi isk/r} \left| x(x^k \mod N) \mod N \right\rangle$$
$$= \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi isk/r} \left| x^{k+1} \mod N \right\rangle$$
$$= \frac{1}{\sqrt{r}} \cdot \underbrace{\frac{e^{2\pi is/r}}{e^{2\pi is/r}}}_{1} \sum_{k=0}^{r-1} e^{-2\pi isk/r} \left| x^{k+1} \mod N \right\rangle$$
$$= \frac{e^{2\pi is/r}}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi is(k+1)/r} \left| x^{k+1} \mod N \right\rangle$$

The sum with $k+1$ is equal to the sum with k because the last term $x^{k+1} = x^r = 1 \mod N$ is equal to the first term $x^k = x^0 = 1$.

$$\frac{e^{2\pi is/r}}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi is(k+1)/r} \left| x^{k+1} \mod N \right\rangle$$
$$= \frac{e^{2\pi is/r}}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi isk/r} \left| x^k \mod N \right\rangle = e^{2\pi is/r} |u_s\rangle$$

Finally, $|u_s\rangle$ is an eigenvector of U_x with eigenvalue $e^{2\pi is/r}$.

7.1.2 Calculating the order

With the eigenvector $|u_s\rangle$ we can use QPE to compute the phase of $e^{2\pi is/r}$ that is $\frac{s}{r}$. Since $\frac{s}{r}$ is a fraction, if we know the maximum value of r we can surely represent it with a finite number of bits.

To use the QPE algorithm we need to prepare an eigenvector as input and this is not trivial, but there is a solution.

Sum of eigenvector for modular multiplication

Given all the eigenvectors $|u_s\rangle$ for the operator U_x we have that:

$$\frac{1}{\sqrt{1}} \sum_{s=0}^{r-1} |u_s\rangle = |1\rangle$$

Proof:

$$\frac{1}{\sqrt{1}} \sum_{s=0}^{r-1} |u_s\rangle = \frac{1}{\sqrt{1}} \sum_{s=0}^{r-1} \frac{1}{\sqrt{r}} \sum_{k=0}^{r-1} e^{-2\pi isk/r} \left| x^k \mod N \right\rangle$$
$$= \frac{1}{r} \sum_{s,k=0}^{r-1} e^{-2\pi isk/r} \left| x^k \mod N \right\rangle$$

If we consider the term of the inner sum where $k = 0$, we have that $x^k = 1$ and $e^{-2\pi isk/r} = 1$, so we can divide that term.

$$\begin{aligned}
&\frac{1}{r}\sum_{s,k=0}^{r-1} e^{-2\pi isk/r}\left|x^k \mod N\right\rangle \\
&= \frac{1}{r}\sum_{s=0}^{r-1}|1\rangle + \frac{1}{r}\sum_{s=0;k=1}^{r-1} e^{-2\pi isk/r}\left|x^k \mod N\right\rangle \\
&= |1\rangle + \frac{1}{r}\sum_{k=1}^{r-1}\left|x^k \mod N\right\rangle\sum_{s=0}^{r-1} e^{-2\pi isk/r} \\
&= |1\rangle + \frac{1}{r}\sum_{k=1}^{r-1}\left|x^k \mod N\right\rangle\sum_{s=0}^{r-1}(e^{-2\pi ik/r})^s \\
&= |1\rangle + \frac{1}{r}\sum_{k=1}^{r-1}\left|x^k \mod N\right\rangle\frac{1-(e^{-2\pi ik/r})^s}{1-e^{-2\pi ik/r}}
\end{aligned}$$

The sum is 0 because $e^{-2\pi ik} = 1$ so the result is simply $|1\rangle$.
We now know that using $|1\rangle$ as the bottom input in QPE will calculate all the phases simultaneously. Since the bottom input of QPE must be L bits long, $|1\rangle$ is defined as $|00\ldots01\rangle$.
Using QPE we have an accurate estimation for $\frac{s}{r}$, to obtain r we can use continued fractions:

$$r = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{a_n}}}}$$

where $a_0, a_1, \ldots, a_n$ are positive integer.

7.2 Shor algorithm

Fundamental theorem of Arithmetic

Any $n \in N$ can be written as:

$$n = p_1^{\alpha_1} \cdot p_2^{\alpha_2} \cdot \ldots \cdot_m^{\alpha_m}$$

where $p_1, \ldots, p_m$ are prime numbers.

The **integer factoring problem** asks to find the primes for a given number n. It suffices for the algorithm to find one prime at a time, because we can recursively call it until we have all factors.

Finding factor by solving an equation

Let N be a L-bit composite number and x be a non-trivial (not 1 or -1) solution to the equation $x^2 = 1 \mod N$, then at least one of $\mathrm{GCD}(x-1, N)$ and $\mathrm{GCD}(x+1, N)$ is a factor of N.

The factor can be computed using $O(L^3)$ operations.

Reducing factorization to order-finding

Let N be a odd composite number with m prime factors and x be an integer chosen at random in the range $[1, N-1]$ and coprime to N. With r order of x modulo N we have that:

$$P(r \text{ is even and } x^{r/2} \neq -1 \mod N) \geq 1 - 2^{-m}$$

For any value of m this probability is always higher than 50%.

Using this two theorem Shor crated its algorithm:

Algorithm: Shor algorithm

```
def Shor(N):
    if N is even :
        return 2
    if N = a^b for a ≥ 1 and b ≥ 2 :
        return a
    x=rand(1,N-1)
    if GCD(x, N) > 1 :
        return GCD(x, N)
    r=order_finding(x,N)
    if r is even and x^{r/2} ≠ −1 mod N :
        return GCD(x^{r/2} − 1, N) or GCD(x^{r/2} + 1, N)// the non-trivial one
```

Considering $L = \lceil \log N \rceil$ as the input lenght, the QPE used in order-finding has complexity $O(L^3)$ and the continued fraction calculation to find r has complexity $O(L^3)$. Using faster multiplication algorithms, the complexity can be reduced to $O(L^2 \log L)$.

8

Quantum Key Distribution

8.1 Privacy Amplification

In private-key criptography two parties (Alice and Bob) must share a private key to encrypt and decrypt their messages, the key must have the same lenght of the message. The difficult part is how to share the key without another person (Eve) intercepting it.

Starting from a bitstring shared between Alice and Bob, from which Eve has some knowledge, they can distil a more secure key that reduces Eve's knowledge. This is done by using a randomly chosen hash function $g : A \to B$ where an element of A is a n-bit string and an element of B is a m-bit string for $m < n$. With this random hash function g we have that:

$$\forall a_1, a_2 \in A \quad P(g(a_1) = g(a_2)) \leq \frac{1}{|B|}$$

Knowledge of Eve after privacy amplification

Let W be the shared bitstring for which Eve has knowledge $> d$ and $S = g(W)$, it can be shown that the knowledge of Eve on S is:

$$\text{knowledge on } S \geq m - 2^{m-d}$$

8.2 BB84 protocol

BB84 is a quantum key distribution protocol that enables Alice and Bob to share the key over public channels.

The main idea of the protocol is the fact that trying to distinguish non-orthogonal states implies disturbance. If we consider a qubit that is either in state $|0\rangle$ or $|1\rangle$, we can know its state precisely by measuring it. The same works for a qubit that is either in state $|+\rangle$ or $|-\rangle$, after we rotate it we can measure precisely its state. But if we have a qubit that is either $|0\rangle$, $|1\rangle$, $|+\rangle$ or $|-\rangle$ we cannot know its state with only one measurement.

Given two non-orthogonal states $|\psi\rangle$ and $|\varphi\rangle$, to distinguish the states Eve must apply a unitary U on a state $|x\rangle$:

$$|\psi \otimes x\rangle = |\psi \otimes y\rangle$$
$$|\varphi \otimes x\rangle = |\varphi \otimes y'\rangle$$

To distinguish the states $|y\rangle$ and $|y'\rangle$ must be different, however U is unitary so:

$$\langle\psi\otimes x|\varphi\otimes x\rangle = \langle\psi\otimes y|\varphi\otimes y'\rangle \implies \langle\psi|\varphi\rangle\langle x|x\rangle = \langle\psi|\varphi\rangle\langle y|y'\rangle$$
$$\implies \langle x|x\rangle = \langle y|y'\rangle = 1 \implies |y\rangle = |y'\rangle$$

So, Eve cannot gain any information.
The protocol works following this steps, considering T the lenght of the message:

1. Alice chose two random bitstring a,b of lenght N and produces N qubits:

$$Q = \bigotimes_{i=0}^{N} |q_{a_i b_i}\rangle$$

   The qubits are encoded in four ways:

$$|q_{00}\rangle = |0\rangle \qquad |q_{10}\rangle = |1\rangle$$
$$|q_{01}\rangle = |+\rangle \qquad |q_{11}\rangle = |-\rangle$$

   Alice sends Q to Bob using a public quantum channel, it can be done using teleportation.

2. Bob now has N qubits:

$$Q' = \bigotimes_{i=0}^{N} |q_{a_i b_i}\rangle$$

   Bob choose a random bitstring b of lenght N and measures each qubit with basis $\{|0\rangle, |1\rangle\}$ or basis $\{|+\rangle, |-\rangle\}$ depending on b' and stores the results in a bitstring a'.

3. Alice and Bob send b and b' respectively.

4. Alice stores in W the bits of a for which $b = b'$ and Bob does the same in W'. They both select a random substring from W or W' and sens them on the channel.

5. If the two substring are equal then they both do a privacy amplification on the remaing bits of W or W' to obtain an highly secure key.

Since a lot of bits of the key are lost during the protocol, the value of N must be a lot larger than T to be sure that there are at least T bits in the final key.

8.3 E91 protocol

8.3.1 Principle of Deferred Measurement

In a quantum circuit, measurements that control operations can always be moved to the end of a circuit. Any controlled operator can be replaced by a quantum controlled operator.



It can be shown that the two circuits are equal.

8.3.2 Measurement in non-diagonal bases

From the spectral theorem shown in Subsection 4.3.1 we know that the set of all eigenvector of a self-adjoint operator acting on a Hilbert space $\mathcal{H}$ is an orthonormal basis for $\mathcal{H}$.

Changing the basis

Given two orthonormal bases $\{e_i\}$ and $\{f_i\}$ we can change bases from $\{f_i\}$ to $\{e_i\}$ using an operator that satisfies:

$$\forall i \quad |f_i\rangle \to |e_i\rangle$$

This can be obtained by using the operator:

$$U = \sum_{\forall i} |e_i\rangle \langle f_i|$$

Example:
To measure a qubit in the basis $\{|+\rangle, |-\rangle\}$ we map $|+\rangle$ to $|0\rangle$ and $|-\rangle$ to $|+\rangle$ using the operator:

$$U = |0\rangle \langle +| + |1\rangle \langle -|$$

The operator U is unitary since:

$$U^\dagger = |+\rangle \langle 0| + |-\rangle \langle 1|$$

We can show:

$$\begin{gathered} UU^\dagger = (|0\rangle \langle +| + |1\rangle \langle -|)(|+\rangle \langle 0| + |-\rangle \langle 1|) \\ = |0\rangle \langle +|+\rangle \langle 0| + |0\rangle \langle +|-\rangle \langle 1| + |1\rangle \langle -|+\rangle \langle 0| + |1\rangle \langle -|-\rangle \langle 1| \\ = |0\rangle \langle 0| + |1\rangle \langle 1| = I \end{gathered}$$

If we write U using matrix representation:

$$\begin{gathered} U = |0\rangle \langle +| + |1\rangle \langle -| = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 1 & 1 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 0 \\ 1 & -1 \end{pmatrix} \\ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = H \end{gathered}$$

In fact the operator the change the basis from $\{|0\rangle, |1\rangle\}$ to $\{|+\rangle, |-\rangle\}$ is H.

8.3.3 Bell Inequalities

Suppose that Alice has two observables Q and R and Bob has other two observables S and T. All observables always return ± 1.
We create an EPR pair like $\frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ and give both Alice and Bob one particle. What happens if Alice and Bob measure a random observable of the two they have?
There are some assumptions:

1. Neither Alice or Bob decide which observables they will measure before recieving their particle.

2. Both Alice and Bob measure their particle using the chosen observable at the same time.

Let us consider the quantity $QS + RS + RT - QT$, its easy to see that:

$$QS + RS + RT - QT \leq 2$$

Bell inequality

Given the four observables Q,R, S and T it hold that:

$$\mathcal{A} = E[QS + RS + RT - QT] \leq 2$$

Since the expected value is linear:

$$\mathcal{A} = E[QS] + E[RS] + E[RT] - E[QT] \leq 2$$

Using as EPR pair $\psi = \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$ with observables:

$$Q = Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \qquad R = X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$
$$S = \frac{-Z - X}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} \qquad T = \frac{Z - X}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}$$

We can calculate expected value of the entangled state:

$$E[QS] = \langle QS \rangle = \langle \psi | (Q \otimes S) | \psi \rangle$$

Changing the values of Q and S we have:

$$\langle \psi | (Q \otimes S) | \psi \rangle = \left\langle \psi \middle| \left(Z \otimes \frac{-Z - X}{\sqrt{2}} \right) \middle| \psi \right\rangle$$
$$= \left\langle \psi \middle| - \left(\frac{Z \otimes Z}{\sqrt{2}} \right) \psi \right\rangle + \left\langle \psi \middle| - \left(\frac{Z \otimes X}{\sqrt{2}} \right) \psi \right\rangle$$
$$= \left\langle \psi \middle| - \left(\frac{Z \otimes Z}{\sqrt{2}} \right) \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \right\rangle + \left\langle \psi \middle| - \left(\frac{Z \otimes X}{\sqrt{2}} \right) \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle) \right\rangle$$
$$= -\frac{1}{\sqrt{2}} \left\langle \psi \middle| \frac{1}{\sqrt{2}}(-|01\rangle + |10\rangle) \right\rangle - \frac{1}{\sqrt{2}} \left\langle \psi \middle| \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \right\rangle$$
$$= \frac{1}{\sqrt{2}} \langle \psi | \psi \rangle + 0 = \frac{1}{\sqrt{2}}$$

The second part is 0 since the two vector are othogonal.
Using some similar calculations we can find:

$$E[QS] = \frac{1}{\sqrt{2}} \qquad E[RS] = \frac{1}{\sqrt{2}}$$
$$E[RT] = \frac{1}{\sqrt{2}} \qquad E[QT] = -\frac{1}{\sqrt{2}}$$

And doing the inequality calculation the result is:

$$\mathcal{A} = \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} - \left(\frac{1}{\sqrt{2}} \right) = 2\sqrt{2} > 2$$

The Bell inequality is violeted by quantum mechanics.

The **E91** protocol uses this idea: Alice and Bob use a random part of their qubits to compute the value of $\mathcal{A}$. If the value of $\mathcal{A} \approx 2\sqrt{2}$ then they can generate keys like in BB84, otherwise they retry.